Fiscal Year:	FY 2024	**Task Last Updated:**	FY 12/31/2023
PI Name:	Tabassum, Shawana Ph.D.		
Project Title:	Leaf Sensor Network for In Situ and Multiparametric Analysis of Crop Stressors		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2021 Space Biology NNH21ZDA001N-SBPS E.9: Plant Studies
Start Date:	03/01/2023	**End Date:**	02/29/2024
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA KSC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Rajan, Nithya Ph.D. (Texas A&M University)		
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	This project aims to advance the fundamental understanding of the hormonal responses in plants in a spaceflight-like environment through an in situ technology that collects and analyzes data on plant phytohormones in real time. Progressive variations in the phytohormone levels initiate a plant's defense mechanisms against environmental stressors. Salicylic acid, jasmonic acid, abscisic acid, and indole-3-acetic acid are among the most important regulators of induced defense mechanisms. Progressive variations in their levels have been reported under many drought and cold/heat-stressed conditions on Earth. However, the dynamic interaction mechanism of these hormones is not fully elucidated in a "space farming" setting due to the lack of technology, needed to facilitate in situ sensing. Real-time understanding of a plant's responses to stressors is essential to minimize stress-induced growth and yield declines in plants. Toward this end, this project proposes to develop a lightweight and integrated plant sensor with sensing elements		

Task Description:	to monitor plant hormonal variations in real-time. The system will comprise a multiplexed hormone sensor for quantitatively measuring the primary defense hormones: Salicylic acid and abscisic acid. The impact of the following stressors on the hormone levels will be analyzed: changes in carbon dioxide (CO2) levels, temperature, and growth media. In contrast to the traditional discrete, disruptive, in vitro, time-intensive, and heavyweight instruments used for molecular analysis, our proposed sensor is plant-wearable and provides in situ monitoring capabilities. We will develop functional correlations of the measured hormonal variations with physiological indicators (photosynthesis, respiration, and transpiration) to differentiate the effect of growing conditions on individual plant productivity during various growth stages. The knowledge gained from this project will advance future research on predicting and improving plant growth and productivity under spaceflight stressors.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This study aims to enhance our understanding of a plant's defense mechanisms against a combination of environmental stressors by providing real-time data on variations in the plant's defense hormones. Accurate and real-time measurements of plant parameters are expected to provide agricultural producers with immediate insights into crop stress levels, facilitating precise adjustments in irrigation and fertilizer application. The outcomes of this project are poised to have a lasting positive impact on terrestrial agriculture by: (1) providing a tool for early assessment of crop water and nutrient uptake, (2) contributing to the existing and new databases on irrigation and fertilizer management, modeling data, and linking local data with national-level databases, (3) developing science-based irrigation and fertilizer management policies, and (4) improving the environmental sustainability of agricultural practices.
Task Progress:	Our central hypothesis is that real-time data collected on the phytohormone levels in plants can be used to determine the fundamental ways plants interact with microclimatic conditions associated with living in space. The expected deliverables of this research are the development of technology and a dataset that could then be used to monitor plant responses and identify the most productive growing conditions in space. In this regard, we have made the following progress and accomplishments during the reporting period. We have developed a three-electrode based electrochemical sensor for salicylic acid detection in plant sap. The sensor design is devised considering that cowpea possesses a dicot stem, leading to the arrangement of vascular bundles along the periphery. Also, the diameter of cowpea plants ranges from 5mm to 10mm. Consequently, the active region of the sensor penetrates through the vascular bundle, leaving the remaining portion accessible for signal acquisition. The sensor was characterized for different concentrations of salicylic acid (SA) typically found in small plants. Like animals, plants produce hormones as signaling molecules to control various growth processes. Salicylic acid (SA) and abscisic acid (ABA) are two primary stress hormones in plants. Progressive variations in their levels have been reported in many drought, salt, and cold/heat-stressed plants. The sensor was calibrated with different SA concentrations. Multiple sensors were calibrated and all of them demonstrated similar responses, therefore confirming their repeatability. The sensor was tested for potential cross-interference from other hormones that are typically present in the sap. The results show that the sensor exhibits minimal response to interfering hormones, when compared to even a very low concentration of SA. We also conducted experiments to assess the reproducibility and longevity of the sensor. The sensor demonstrated reproducible response for up to 3 days. Beyond this period, the sensor's performance began to decline, a phenomenon attributed to the degradation of the immunoassay. To overcome this challenge, our subsequent efforts will focus on substituting the assay with a more durable assay to prolong the sensor's functional lifespan. We tested the growth facilities at Texas A&M University and tested the growth of cowpea under controlled growth chamber conditions in coir pith, two types of hydroponic solutions, and a mixture of peat and perlite. We observed that plants grown in hydroponics had the highest stomatal conductance and transpiration, which indicates high water use. The experiments involving the sensors under two CO2 levels and temperature conditions will commence in February 2024. Our immediate research plan includes the following: • replace the current immunoassay with a more durable assay to prolong the sensor's functional lifespan • develop and optimize an abscisic acid sensor alongside the SA sensor • test the hormone sensors in cowpea plants under the effects of growth medium and elevated CO2 and temperature levels. These tests to identify the correlations between hormonal variations and plant physiological responses will commence in February 2024. This experiment will use cowpea as the test plant due to its ultra-short duration life cycle and versatility of use. Plants will be grown in a super lightweight media of coir pith and compared against hydroponics and regular potting mix, which will be used as a control. The sensors will be deployed 20 days after plant germination and plant performance will be monitored. Plants will be grown under ambient CO2 conditions and elevated CO2 conditions in growth chambers located at Texas A&M University. NASA facilities will not be used for the plant growth experiments.
Bibliography Type:	Description: (Last Updated: 08/06/2025)
Articles in Peer-reviewed Journals	Hossain NI, Noushin T, Tabassum S. "Tattoo-like flexible ethylene sensor for plant stress monitoring in real-time." IEEE Sens J. 2023 Oct 31;1. http://dx.doi.org/10.1109/JSEN.2023.3327547 , Oct-2023